The **ilium** is the superior flaring portion of the hip bone. Each ilium forms a secure joint with the sacrum posteriorly. The **ischium** is a curved bar of bone. The V-shaped **pubic bones** articulate anteriorly at the **pubic symphysis**. The pelvis of male is deep and narrow with larger heavier bones and the female is shallow, wide and flexible in nature, and this helps during pregnancy which is influenced by female hormones.

The pelvic girdle is a heavy, strong girdle. How does its structure reflect its function?

The Lower limb

The lower limb consists of 30 bones which carries the entire weight of the erect body and is subjected to exceptional forces when we jump or run. The bones of the lower limbs are thicker and stronger than the upper limbs. The three segments of each lower limb are **the thigh**, **the leg** or **the shank** and **the foot**. **The femur** is the single bone of the thigh. It is the largest, longest and strongest bone in the body. The head of femur articulates with **the acetabulum** of the pelvis to form the hip joint. Two parallel bones, **the tibia** and **fibula**, form the skeleton of the shank. A thick, triangular **patella** forms the knee cap, which protects the knee joint anteriorly and improves the leverage of thigh muscles acting across the knee. The foot includes the bones of ankle, **the tarsus**, **the metatarsus** and **the phalanges or toe bones**. The foot supports our body weight and acts as a lever to propel the body forward, while walking and running. **The tarsus** is made up of seven bones called tarsals. **The metatarsus** consists of five bones called metatarsals. The arrangement of the metatarsals is parallel to each other. There are 14 **phalanges** in the toes which are smaller than those of the fingers.

Structure of a typical long bone

A typical long bone has a **diaphysis**, **epiphyses** (singular-epiphysis) and **membranes** (Figure 9.10). A tubular diaphysis or shaft, forms the long axis of the bone. It is constructed of a thick collar of compact bone that surrounds a central **medullary cavity** or **marrow cavity**. The epiphyses are the bone ends. Compact bone forms the exterior of epiphyses and their interior contains spongy bone with red marrow. The region where the diaphysis and epiphyses meet is called the **metaphysis**. The external surface of the entire bone except the joint surface is covered by a double-layered membrane called the **periosteum**. The outer fibrous layer is dense irregular connective tissue. The inner osteogenic layer consists of **osteoblasts** (bone- forming cells) which secrete bone matrix elements and **osteoclasts** (bone-destroying cells). In addition, there are primitive stem cells, osteogenic cells, that give rise to the osteoblasts. The periosteum is richly supplied with nerve fibres, lymphatic vessels and blood vessels. Internal bone surfaces are covered with a delicate connective tissue membrane called the **endosteum**. The endosteum covers the trabeculae of spongy bone and lines the canals that pass through the compact bone. It also contains both osteoblasts and osteoclasts. Between the epiphysis and diaphysis **epiphyseal plate** or **growth plate** is present.

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